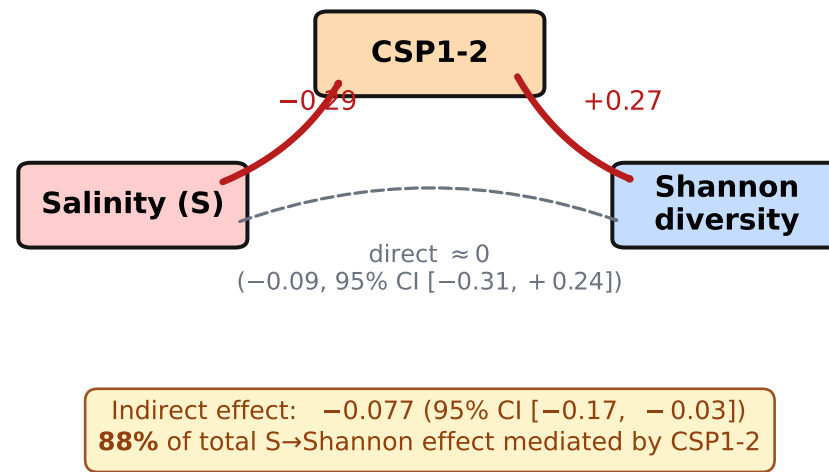
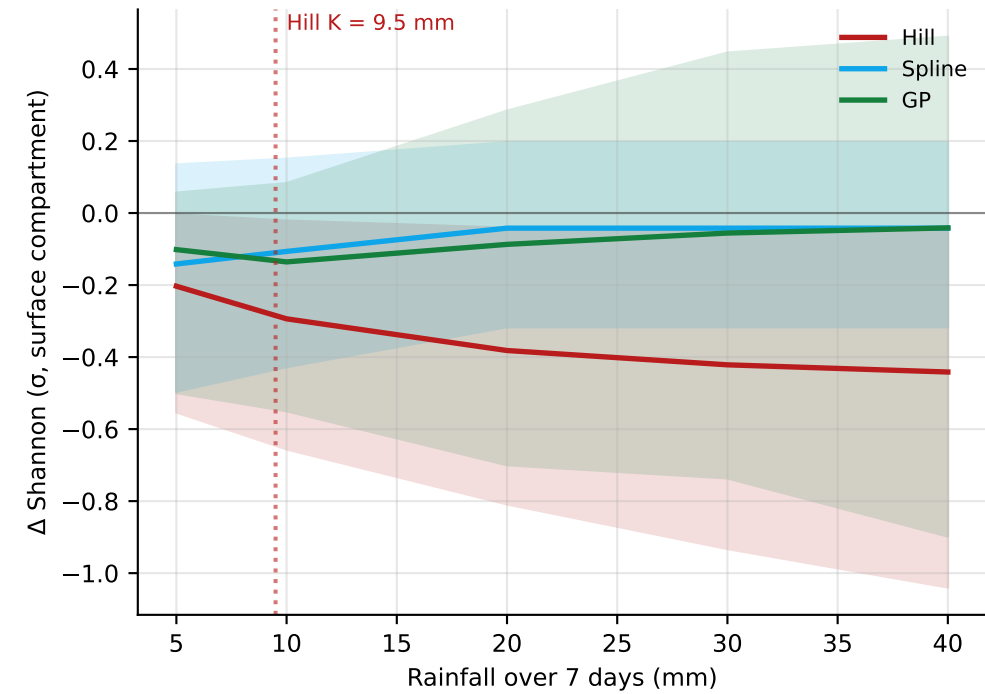


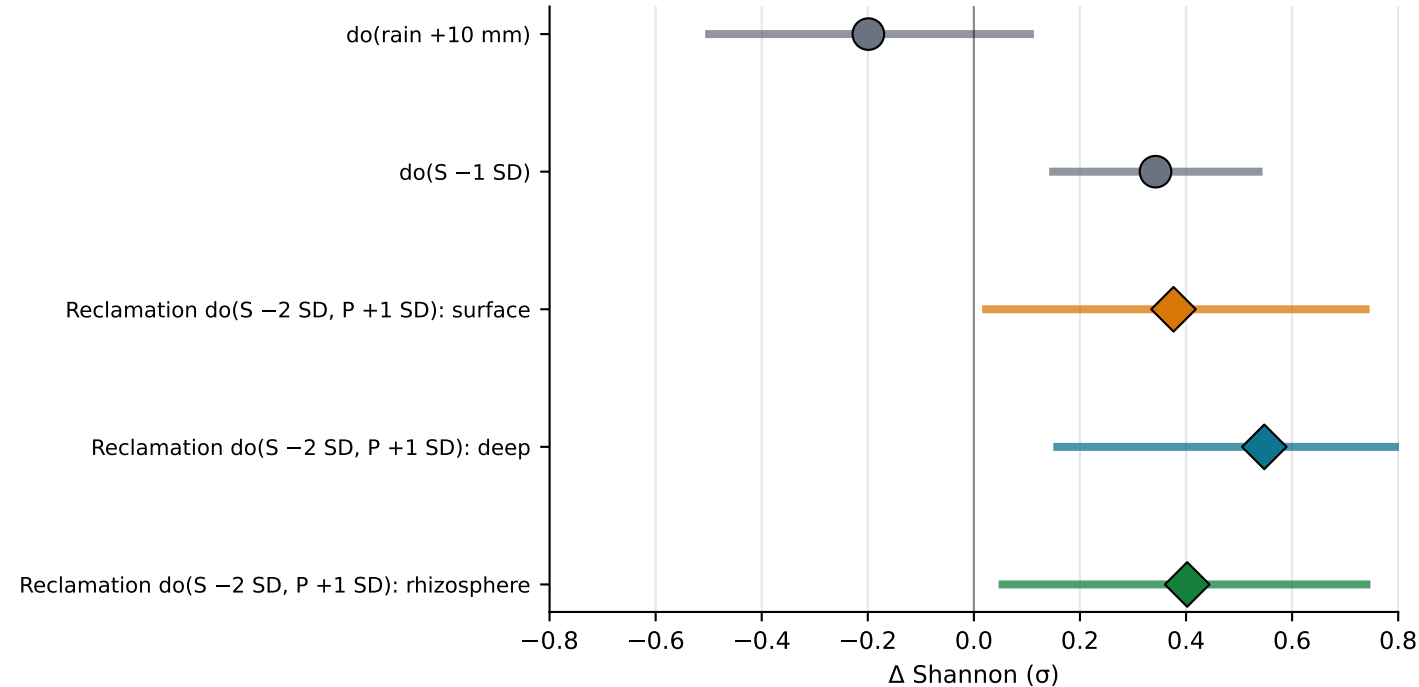
(a) Causal mediation: salinity $\rightarrow$ CSP1-2 $\rightarrow$ Shannon



(b) Non-linear rainfall dose-response (Hill / spline / GP)



(c) Interventional posteriors (Bayesian state-space digital twin)



(d) JSMD knockout — top-affected genera
 Δ Shannon ≈ -0.35 propagates through biotic associations

